CSE 219 — Signals and Linear Systems

Exam Suggestion · Anik Sir's Part (Lectures 1–7)

Syllabus rules · What to study · Likely questions · Short proofs · How to write (Anik Sir's portion)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

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> **NOTE** Built from Anik Sir's syllabus message and from the 2022-23 and 2023-24 CSE 219 papers, plus the earlier EEE 211 / CSE 311 papers on the same content. Lesson numbers (L1–L19) refer to CSE219_Anik_*_Notes.pdf. This is a suggestion, not a guarantee: the questions below are the types most likely given the slides and the pattern.

1. Syllabus Rules from Anik Sir

- **Read all slides, Lectures 1–7, carefully.** That is enough. Questions are mathematical problems, proofs, or application-based theory.
- **Not needed for the exam:** very long proofs that are the foundation of a theory — how the Fourier transform comes from the Fourier series (Lec 7, sl. 4–18), how convolution and the Fourier-series equation are derived. Read them for intuition only; do not memorise them to write.
- **Excluded:** every slide whose title is marked **(*)** — in Lec 2 the smoothed-step derivation of δ(t) (sl. 20–25) and in Lec 5 the coordinate/projection intuition (sl. 44–52).
- **Excluded:** Parseval's theorem (taught briefly in one class, not in Part 1). Skip the Parseval questions in the old papers (e.g. 2023-24 Q2(b)).
- **Short proofs are in** (see section 4): even–odd decomposition, impulse-train transform is a train, the stability condition, and similar small ones.
- **Application questions** will be like the labs: where and how convolution, Fourier series and the Fourier transform are used. The practice sheet has sample problems.
- Ask Anik Sir if anything is unclear.

> **NOTE** Because the notes were written from the whole TF collection, a few parts of them go beyond the slides: orthogonal functions and Dirichlet conditions (L11), the Fourier-series symmetry rules and Parseval (L13, L15), the transform-pair table beyond the rectangle/impulse and Parseval (L18), filters and bandwidth (L19). Treat those as **low priority** — read them only after everything below.

2. How the Paper Is Likely to Look

In 2023-24 Anik Sir's part was Section A: **Q1 compulsory (45 marks, three parts of 15)** and **two of Q2–Q4 (30 each)**. 2022-23 had the same shape (Q5 compulsory 45; two of Q6–Q8). Expect:

Slot	What it was	What to expect this time
Q1(a), 15	2023-24: sketch $x(-2(t-1)) + x((t+2)/2)$. 2022-23: even–odd decomposition of a piecewise signal	Transformation sketch of a piecewise signal (shift, reversal, scale in the right order) **or** even–odd parts with graphs
Q1(b), 15	2023-24: DT LTI interconnection, h_eq and y[n]. 2022-23: inverse FT chain with properties	Interconnected LTI systems / convolution; or a short chain using FT properties
Q1(c), 15	2023-24: FT of rect(t), then $\text{rect}((t-\tau)/T)\cos 2\pi f_0 t$. 2022-23: periodicity and FS check	FT of a simple signal, then a property (shift, modulation); or periodicity of a sinusoid
Q2–Q4, 30 each	2023-24: FT of $e^{-at}u$ and $e^{-a\|t\|}$; DT periodicity + causal/stable; odd square wave with a shift. 2022-23: FT of f(t), sign(t); $x(-2t+2)$ by properties; 2-sided DT convolution	One FT-properties problem; one convolution (DT or CT, graphical) with causal/stable; one Fourier-series problem; plus one application/theory part inside a question

3. Lecture-by-Lecture Priorities

Lecture	Study (★★★ must, ★★ should)	Practice from
Lec 1 — Signals (L1–L3)	★★★ x(αt+β) recipe: shift by β, reverse if α<0, scale by |α|; write the intermediate signal each time (sl. 26–39). ★★★ Even–odd parts: x_e = ½[x(t)+x(−t)], x_o = ½[x(t)−x(−t)] and the graphs (sl. 40–52). ★★ Energy and power, E and P of the half-sine, periodic signals, 1/|t| signal with finite energy (sl. 6–16).	Practice-sheet 1.1–1.8 (esp. 1.5, 1.6, 1.7, 1.8). Notes L2 Q7 and 2023-24 Q1(a); L3 2022-23 Q5(a).
Lec 2 — Useful signals (L4, L5)	★★★ DT periodicity: $2\pi/\omega_0$ must be rational (cos 2n no, cos(2πn/5) yes, 2cos(16πn/63+π/3) N = 63) (sl. 43–50). ★★ Step/impulse: δ[n] = u[n]−u[n−1], u[n] = Σδ[m] (sl. 11–15); jumps → impulses in the derivative; $e^{-t}u(t-2)$, e^tu(3−t) (sl. 31–35). Skip the (*) slides 20–25.	Practice-sheet 2.1–2.6. Notes L5 Q(a)–(c) (periodicity of x[n] = $(3-4j)e^{j13\pi n/7}$); L4 jump-signal derivative.
Lec 3 — LTI and convolution (L6–L8)	★★★ Linearity and time invariance tests; impulse response; y[n] = Σx[k]h[n−k] by reflect–shift–multiply–add (sl. 13–28, 47–57). ★★ Convolution = polynomial multiplication; echo and delay systems h[n] = u[n], $\alpha^n u[n]$, δ[n−3]. Understand, do not memorise, the derivation of y = x*h.	Practice-sheet 3.1–3.3. Notes L6 and L7 practice; 2023-24 Q1(b) (interconnection) in L9.
Lec 4 — Convolution, causality, stability (L9, L10)	★★★ DT and CT convolution of one-sided exponentials with the overlap/case split ($\alpha^n u$*u, 2^n u[−n]*u, $e^{-at}u$*u, e^tu(−t)*$e^{-t}u$) (sl. 11–52). ★★★ Causal ⇔ h = 0 for t<0; stable ⇔ ∫|h| < ∞; do $h_1 = e^{2t}u(-2-t)$ and $h_2 = te^{-t}u(t)$ (sl. 55–69). ★★ A periodic h cannot be causal or stable.	Practice-sheet 4.1–4.8. Notes L9 Q(a) (2022-23 Q8(a), 20 marks) and L10 (b), (a).
Lec 5 — Fourier series (L11, L12)	★★★ Coefficients by inspection (Euler) and by integration: a_0 = average, $a_k = (1/T)\int x\, e^{-jk\omega_0 t}dt$ (sl. 18–29). ★★★ Square wave: $a_0 = 2T_1/T$, $a_k = \sin(k\omega_0 T_1)/(k\pi)$, T_1 = T/4 spectrum (sl. 30–34). ★★ Ramp/triangle example (sl. 35–41; the slide prints $6k^2\omega_0^2$ — the correct factor is $2k^2\omega_0^2$). Skip (*) sl. 44–52.	Practice-sheet 5.1–5.6. Notes L12 practice; mean value by area (5.4).
Lec 6 — FS properties (L13, L14, L15 partly)	★★★ Shift $a_k e^{-jk\omega_0 t_0}$, reversal a_{-k}, scaling (same a_k, $\omega_0 \to \alpha\omega_0$), linearity, combined x(4−3t) (sl. 9–35). ★★★ Differentiation $jk\omega_0 a_k$ and impulses at jumps; integration $a_k/(jk\omega_0)$ with a_0 separately; sifting; impulse train $a_k = 1/T$ (sl. 36–72). ★★ Multiplication ↔ convolution of coefficients: $(1+\cos\omega_0 t)^2$ (sl. 73–81).	Practice-sheet 6.1–6.5. Notes L13 Q(a) (odd square wave with shift, 2023-24 Q4), L14.
Lec 7 — Fourier transform (L16, L17, parts of L18)	★★★ Rect ↔ $2\sin(\omega T/2)/\omega$, shifted rect, Lego-block decomposition (sl. 29–38). ★★★ Frequency shift and cosine modulation x cos $\omega_0 t$ ↔ ½X(ω−ω₀)+½X(ω+ω₀); time shift $e^{-j\omega t_0}X$ (sl. 19–28, 35). ★★★ δ(t) ↔ 1, $\delta(t-t_0) \leftrightarrow e^{-j\omega t_0}$ (sl. 41–44). ★★ Applications (sl. 45–59): denoising, JPEG, MRI, music ID, vibration, ECG. Pairs beyond the slides ($e^{-at}u$, sgn, $e^{-a\|t\|}$, duality) are asked in the older papers and in 2022-23/2023-24 — learn them from L18 after the slide material. Skip the FS→FT derivation (sl. 4–18).	Practice-sheet 7.1–7.7. Notes L17 Q(a) (2023-24 Q1(c)), L18 Q(a)–(b) except the Parseval part.

4. The Short Proofs You Must Be Able to Write

Each takes 3–6 lines. Write the statement, the key step, the conclusion.

Proof	Key step
1. Every signal has an even and an odd part (Lec 1, sl. 43–44)	Define $x_e = ½[x(t)+x(-t)]$, $x_o = ½[x(t)-x(-t)]$. $x_e(-t) = x_e(t)$ and $x_o(-t) = -x_o(t)$ by substitution, and $x_e + x_o = x$.
2. Impulse train in time is an impulse train in frequency (Lec 6, sl. 52–56)	Period $-T/2 < t < T/2$ contains one impulse; sifting gives $a_k = (1/T)\int\delta(t)e^{-jk\omega_0 t}dt = 1/T$ for every k — a flat line spectrum, i.e. a train in k (and in ω at $k\omega_0$).
3. Stability condition $\int\|h\| < \infty$ (Lec 4, sl. 58–59)	$\|y(t)\| = \|\int h(\tau)x(t-\tau)d\tau\| \le B\int\|h(\tau)\|d\tau$ for $\|x\| \le B$. So finite area ⇒ bounded output.
4. A periodic h cannot be causal or stable (Lec 4, sl. 60–62)	Causal: a nonzero $h(t_0)$ repeats at $t_0 - kT < 0$. Stable: area A per period × infinitely many periods = ∞.
5. Time shift ↔ phase, frequency shift ↔ modulation, FS time shift (Lec 6 sl. 11–12; Lec 7 sl. 22–25)	$x(t-t_0) = \Sigma a_k e^{jk\omega_0(t-t_0)} = \Sigma[a_k e^{-jk\omega_0 t_0}]e^{jk\omega_0 t}$; FT: substitute $\tau = t-t_0$. Frequency shift: $x e^{j\omega_0 t} \leftrightarrow X(\omega-\omega_0)$ because every $e^{j\omega t}$ component becomes $e^{j(\omega+\omega_0)t}$.
6. Reversal/scaling/differentiation of FS coefficients (Lec 6)	Substitute into $x = \Sigma a_k e^{jk\omega_0 t}$: $x(-t) \to a_{-k}$; $x(\alpha t) \to$ same a_k with $\omega_0' = \alpha\omega_0$; $dx/dt \to jk\omega_0 a_k$.
7. y = x*h for LTI (Lec 3, sl. 52–57) — understand, do not memorise	$x = \Sigma x[k]\delta[n-k]$; TI gives $\delta[n-k]\to h[n-k]$; linearity gives $y = \Sigma x[k]h[n-k]$. Three lines are enough if asked.

TRAP Not expected as full derivations: Fourier series → Fourier transform ($T \to \infty$), the general convolution/FS equations from scratch, and anything about Parseval.

5. Application-Type Questions

These are theory + common sense, like the labs. Prepare two or three sentences for each:

Topic	Say this
Convolution	Output of any LTI system = input * impulse response. Room reverb: convolve the dry voice with the hall's measured impulse response (clap/pop). Echo: h = δ[n] + aδ[n−D] (D = delay, a = strength). Smoothing sensor noise: convolve with a moving-average kernel. Image blur/sharpen: 2-D convolution.
Fourier series	Periodic signal = sum of harmonics $k\omega_0$: solve each easy sine, add the answers (vibrating string, heat diffusion). Square/triangle waves, harmonic analysis of a periodic voltage, AC circuit response harmonic by harmonic.
Fourier transform	Aperiodic signals have a continuous spectrum. Audio/image denoising (remove noise frequencies), JPEG (drop small high-frequency coefficients), MRI (data measured in the frequency domain), music recognition (fingerprint from frequency peaks), vibration diagnosis, ECG 50 Hz notch.
Practice-sheet sample problems	4.9 room acoustics, 4.10 smoothing sensor readings, 4.11 echo, 7.5 hidden signals in space, 7.6 removing a tone, 7.7 audio compression — write 3–4 lines each.

6. How to Do the Maths in the Exam

- **Visualise.** Sketch the signal roughly. Even–odd: draw x(t), x(−t), then x_e and x_o — you will see at once whether x_e is mirror-symmetric and x_o is point-symmetric. Convolution: draw x(τ) and h(t−τ), slide h and watch where they overlap; the interval table follows from the picture.
- **Rough is fine.** The sketch is for you; mark the key points (break points, heights) and move on.
- **Do not write everything formally.** Show the main steps in your own way, with the right sketch and the correct limits; finish in time.
- **For transformation sketches** write the intermediate signal after every operation (the slides ask for it), and check one break point with t = (a−β)/α.
- **For convolution** always give the case split (n<0, 0≤n, …) and check: support = sum of supports, continuity at the boundaries.
- **For causal/stable questions** state the support of h first (causal), then compute ∫|h| (stable).
- **For Fourier-series questions** try inspection first (Euler); integrate only if the signal is a standard waveform; use the properties to avoid a second integral.

7. One-Week Plan

Day	Do
1	Lec 1: transformation recipe (all examples), even–odd; practice-sheet 1.1–1.8
2	Lec 2: periodicity, step/impulse, exponential-step signals; practice-sheet 2.1–2.6; Lec 3 systems and discrete convolution
3	Lec 4: CT/DT convolution cases, causality, stability (all slide examples twice); practice-sheet 4.1–4.8
4	Lec 5: coefficients by inspection and integration; square wave, ramp; practice-sheet 5.1–5.6
5	Lec 6: FS properties (shift/reversal/scaling/derivative/integration/product); practice-sheet 6.1–6.5
6	Lec 7: rect, shifts, modulation, δ; then pairs and properties from L18 ($e^{-at}u$, sgn, $e^{-a\|t\|}$); practice-sheet 7.1–7.4
7	Past CSE 219 papers (2022-23 Q5–Q8, 2023-24 Q1–Q4) timed; the short proofs; application sentences; read the notes' Quick Notes